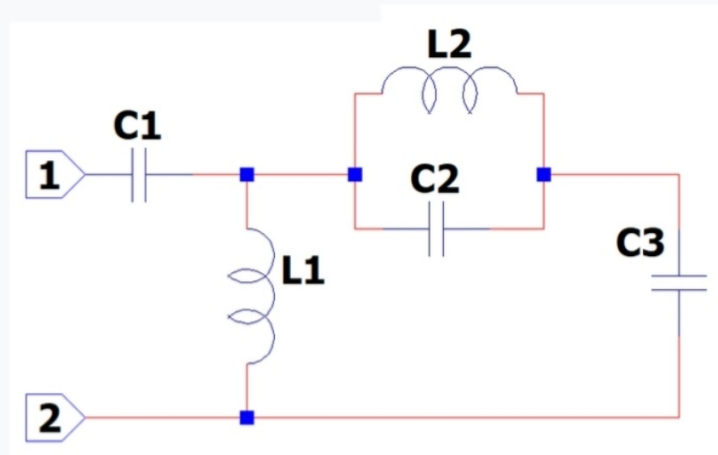


1. Hallar el valor de los componentes sabiendo que satisface la función admitancia propuesta:

$$Y(s) = \frac{s^5 + 18s^3 + 48s}{6s^4 + 42s^2 + 48}$$



$$Z(s) = \frac{6s^4 + 42s^2 + 48}{s^5 + 18s^3 + 48s}$$

$$\begin{aligned} &\rightarrow \frac{48 + 42s^2 + 6s^4}{48 + 18s^2 + s^4} \bigg| \frac{48s + 18s^3 + s^5}{5} \quad \text{--- } C=1 \\ &\quad \frac{48s + 18s^3 + s^5}{24s^2 + 5s^4} \quad \frac{1}{5} \quad \text{--- } L=\frac{1}{2} \\ &\quad \frac{48s + 60s^3 + 0}{24s^2 + 5s^4} \quad 2 \frac{1}{5} \quad \text{--- } L=\frac{1}{2} \\ &\quad \frac{24s^2 + 5s^4}{8s^3 + s^5} \quad \text{--- } Z_2 \end{aligned}$$

$$Z_2 = \frac{24s^2 + 5s^4}{8s^3 + s^5} = \frac{5s^2 + 24}{s^3 + 8s} = \frac{5(s^2 + \frac{24}{5})}{s(s^2 + 8)}$$

→ separo en fracciones (tipo Foster)

$$Z_2 = \frac{5}{5} \frac{(s^2 + \frac{24}{5})}{(s^2 + 8)} = \frac{A}{5} + \frac{Bs}{(s^2 + 8)}$$

$$Z_2 = \frac{1}{sC_2} + \left(\frac{1}{sL} + sC \right)^{-1}$$



$$Z_2 = \frac{1}{sC_2} + \left(\frac{1 + s^2 CL}{sL} \right)^{-1} = \frac{1}{sC_2} + \frac{\frac{s}{C}}{s^2 + \frac{1}{LC}}$$

$$\frac{\left(s^2 + \frac{1}{LC} \right) + \frac{s}{C} \cdot sC_2}{\left(s^2 + \frac{1}{LC} \right) sC_2} \Rightarrow \frac{s^2 \left(1 + \frac{C_2}{C} \right) + \frac{1}{LC}}{\left(s^2 + \frac{1}{LC} \right) sC_2}$$

$$\frac{\left(s^2 + \frac{1}{LC} \frac{1}{1 + C_2/C} \right) \left(1 + \frac{C_2}{C} \right)}{\left(s^2 + \frac{1}{LC} \right) sC_2}$$

$$\begin{cases} \frac{1 + \frac{C_2}{C}}{C_2} = 5 & \rightarrow \frac{1}{C_2} + \frac{1}{C} = 5 \\ \frac{1}{LC} = 8 & \rightarrow \frac{1}{LC} = 8 \\ \frac{1}{LC + C_2 L} = \frac{24}{5} & \rightarrow \frac{1}{LC + C_2 L} \end{cases}$$

→ Para hallar C_2 necesito K_0

$$Z_2 = \frac{5}{s} \frac{(s^2 + \frac{24}{5})}{(s^2 + 8)} = \frac{A}{s} + \frac{Bs}{(s^2 + 8)} \quad K_0 = \frac{5(\frac{24}{5})}{8} = \frac{24}{8}$$

$$\therefore C_2 = \frac{8}{24} = \frac{1}{3}$$

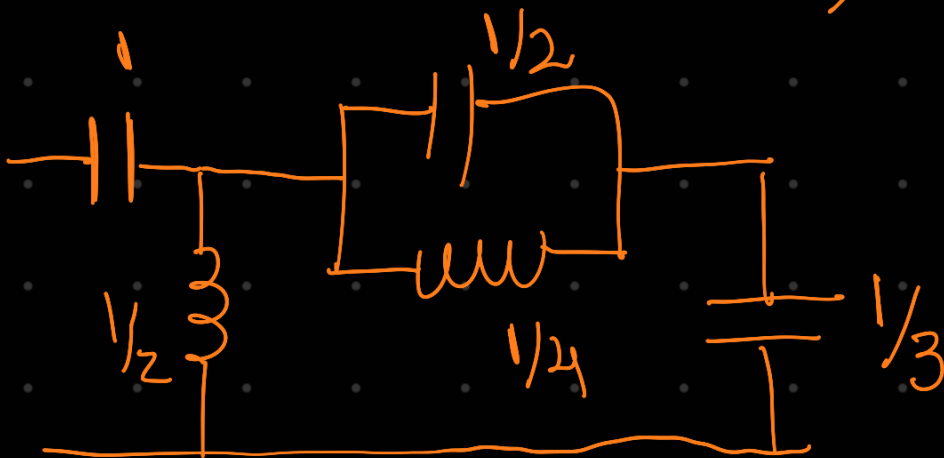
→ Se podría hallar el residuo en -8 que sería igual a $1/c$

$$\left(\frac{s^2 + 8}{s} \right) Z_2 \Big|_{s^2 = -8} \rightarrow \frac{5(-8 + \frac{24}{5})}{-8} = \frac{-40 + 24}{-8}$$

$$K_{-8} = 5 - 3 = 2 \quad \therefore C = 1/2 \quad \checkmark$$

→ Verifiquemos $\frac{1}{C_2} + \frac{1}{C} = 5 \rightarrow \frac{1}{\frac{1}{3}} + \frac{1}{\frac{1}{2}} = 5 \quad \checkmark$

$$\frac{1}{LC} = 8 \rightarrow L = \frac{1}{C8} = \frac{2}{8} = \frac{1}{4} \quad \checkmark$$



A continuación hay síntesis canónicas

→ Usando Caver (an baja freq)

$$\begin{array}{r} 48 + 42s^2 + 65^4 \\ - 48 + 18s^2 + s^4 \end{array} \quad \begin{array}{r} 48s + 18s^3 + s^5 \\ \hline \end{array}$$

$$-||^{C=1}$$

$$\begin{array}{r} 48s + 18s^3 + s^5 \\ - 48s + 18s^3 + s^5 \end{array} \quad \begin{array}{r} 24s^2 + 5s^4 \\ \hline \end{array} \quad \frac{1}{5} \quad \begin{array}{r} 2 \frac{1}{5} \end{array}$$

$$L = \frac{1}{2}$$

$$\begin{array}{r} 24s^2 + 5s^4 \\ - 24s^2 + 3s^4 \end{array} \quad \begin{array}{r} 8s^3 + s^5 \\ \hline \end{array} \quad \frac{1}{5} \quad \begin{array}{r} 1 \frac{1}{3} \end{array}$$

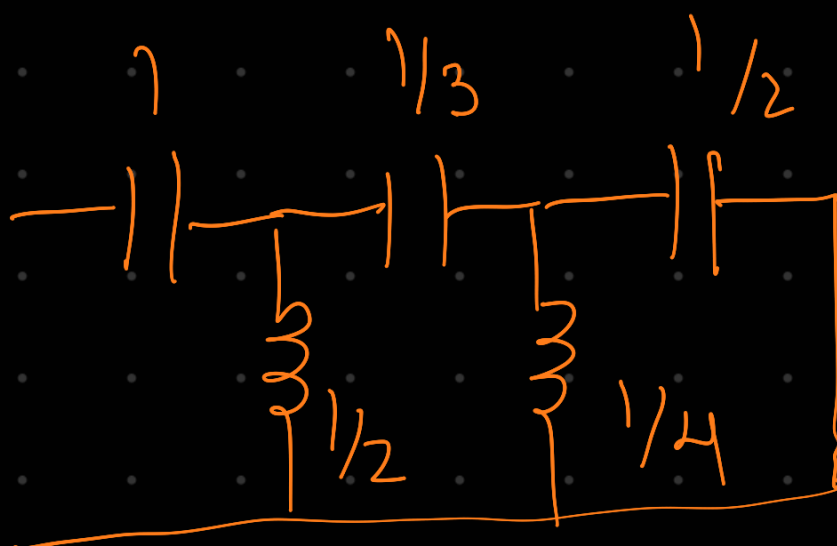
$$-||^{1/3}$$

$$\begin{array}{r} 8s^3 + s^5 \\ - 8s^3 \end{array} \quad \begin{array}{r} 25^4 \\ \hline \end{array}$$

$$\begin{array}{r} 8s^3 \\ - 8s^3 \end{array} \quad \begin{array}{r} 25^4 \\ \hline \end{array} \quad \begin{array}{r} 4 \frac{1}{5} \end{array} \quad \begin{array}{r} 1/4 \end{array}$$

$$\begin{array}{r} 25^4 \\ - 25^4 \end{array} \quad \begin{array}{r} 5^5 \\ \hline \end{array}$$

$$\begin{array}{r} 25^4 \\ - 25^4 \\ \hline 0 \end{array} \quad \begin{array}{r} 2 \frac{1}{5} \end{array} \quad \begin{array}{r} 1/2 \end{array}$$



→ Usando Foster

$$Y(s) = \frac{s^5 + 18s^3 + 48s}{6s^4 + 42s^2 + 48} \quad \rightarrow k_0 \rightarrow 0$$

$$\rightarrow k_\infty \rightarrow 1/6$$

$$Y(s) = \frac{s}{6} \frac{[s^4 + 18s^2 + 48]}{s^4 + 7s^2 + 8} = \frac{s}{6} \frac{(s^2 + \alpha)(s^2 + \gamma)}{(s^2 + z)(s^2 + t)}$$

$$(s^2 + \alpha)(s^2 + \gamma) = s^4 + As^2 + B = s^4 + 18s^2 + 48$$

$$\alpha + \gamma = A = 18$$

$$zt = B = 48$$

$$z = \frac{7 \pm \sqrt{17}}{2}$$

$$\gamma = 9 \pm \sqrt{33}$$

$$t = \frac{16}{7 \pm \sqrt{17}}$$

$$\alpha = \frac{48}{9 \pm \sqrt{33}}$$

$$\therefore Y(s) = \frac{s}{6} \frac{(s^2 + \alpha)(s^2 + \gamma)}{(s^2 + z)(s^2 + t)}$$

$$\cancel{s^5} + 18\cancel{s^3} + 48\cancel{s} = \left[\frac{\cancel{Bs}}{s^2 + z} + \frac{\cancel{Cs}}{s^2 + t} + \frac{\cancel{s}}{6} \right] (s^2 + z)(s^2 + t)6$$

$$s^4 + 18s^2 + 48 = s^4 + s^2[(t+z) + 6B + 6C] + [zt + 6Bt + 6Ct]$$

$$18 = t + z + 6B + 6C$$

$$3 = \frac{t+z}{6} + B + C$$

$$3 - \frac{t+z}{6} - B = C$$

$$48 = zt + 6Bt + 6Cz$$

$$8 = \frac{zt}{6} + Bt + Cz$$

$$\frac{8}{z} - \frac{t}{6} - \frac{Bt}{z} = C$$

igualando

$$B = \frac{187 + 3\sqrt{17}}{204}$$

$$C = \frac{187 - 3\sqrt{17}}{204}$$

$$Y(s) = \frac{s}{6} + \frac{Bs}{s^2 + z} + \frac{Cs}{s^2 + t}$$

$$\frac{1}{s} = \left(sL + \frac{1}{sC} \right)^{-1} \rightarrow \left(\frac{s^2 LC + 1}{sC} \right)^{-1} \rightarrow \frac{sC}{s^2 + \frac{1}{LC}}$$

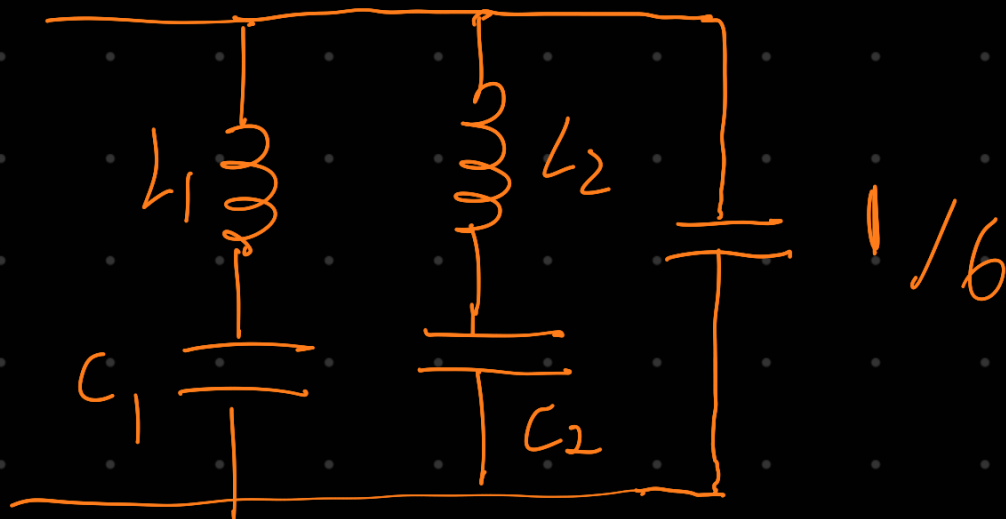
$$B = 2k_i = \frac{1}{L_i}$$

$$\begin{cases} L_1 = 1/B \\ L_2 = 1/C \end{cases}$$

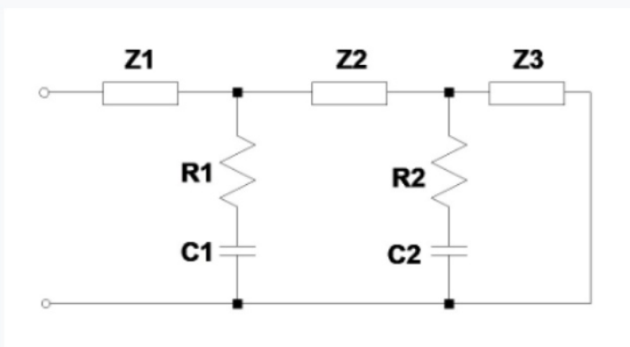
$$\frac{1}{LC_i} = z$$

$$C_i = \frac{1}{zL}$$

$$\begin{cases} C_1 = B/z \\ C_2 = C/t \end{cases}$$



2) Encuentre el valor de los componentes del siguiente circuito:



Sabiendo que está caracterizado por la siguiente función de excitación y constantes de tiempo:

$$Z(s) = \frac{s^2 + 6s + 8}{s^2 + 4s + 3}$$

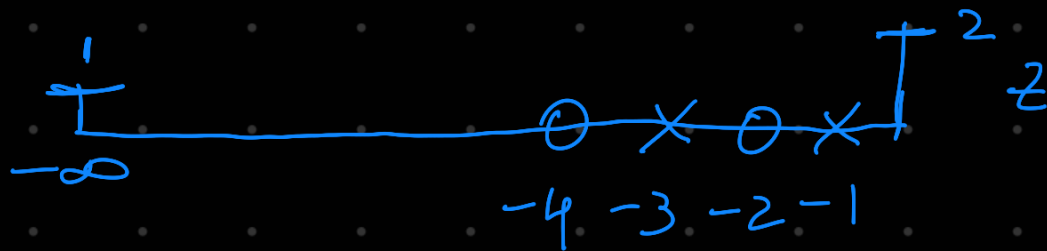
$$R_1 \cdot C_1 = \frac{1}{6} \rightarrow \frac{1}{RC} = 6$$

$$R_2 \cdot C_2 = \frac{2}{7} \rightarrow \tau/2$$

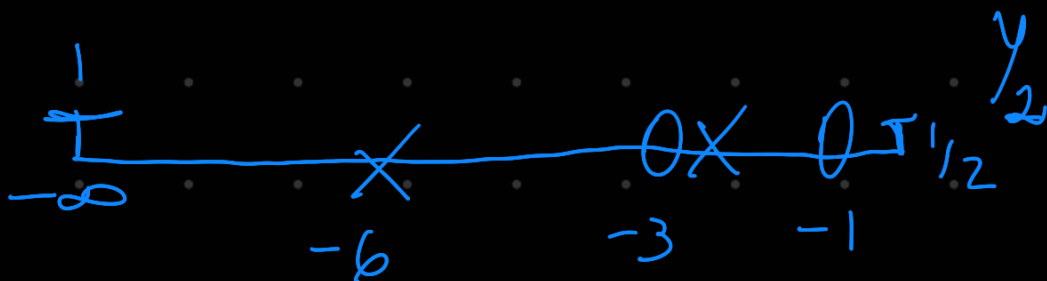
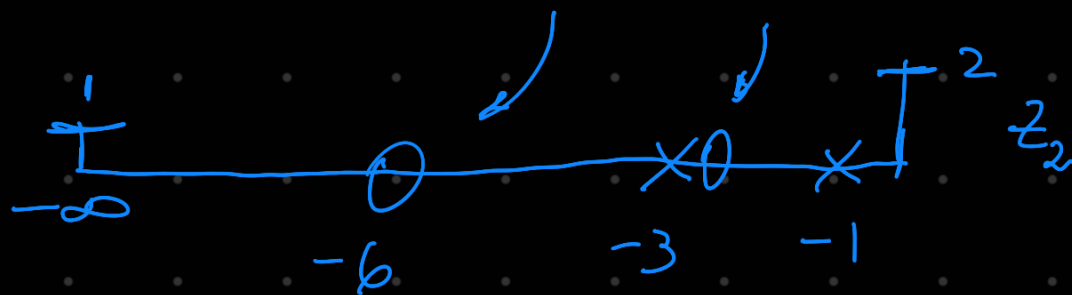
Handwritten notes and formulas:

- For a parallel RC branch: $Y = \left(R + \frac{1}{sC}\right)^{-1}$
- For a series RC branch: $Y = \left(\frac{R s C + 1}{s C}\right)^{-1}$
- For a parallel RC branch: $Y = \frac{3}{R}$
- Partial fraction decomposition: $Z(s) = \frac{s^2 + 6s + 8}{s^2 + 4s + 3} = \frac{(2+s)(4+s)}{(1+s)(3+s)}$
- Factorization: $s + \frac{1}{RC}$ (circled in orange)
- Conclusion: *el polo es en $1/RC$*

$$Z(s) = \frac{s^2 + 6s + 8}{s^2 + 4s + 3} = \frac{(2+s)(4+s)}{(1+s)(3+s)}$$



Quiero setear
cero en -6
para tener
polo en y en
-6 para luego
remover



$$z_2 = z(s) - z(-6)$$

$$z(-6) = \frac{36 - 36 + 8}{36 - 24 + 3}$$

$$z(-6) = \frac{8}{15}$$

Remoción Parcial

$$\frac{8}{15}$$

$$z_2 = \frac{s^2 + 6s + 8}{s^2 + 4s + 3} - \frac{8}{15} = \frac{(s^2 + 6s + 8) - \frac{8}{15}(s^2 + 4s + 3)}{s^2 + 4s + 3}$$

$$z_2 = \frac{\frac{7}{15}s^2 + \frac{58}{15}s + \frac{32}{5}}{s^2 + 4s + 3} = \frac{\frac{1}{15}(16 + 7s)(6 + s)}{(1 + s)(3 + s)}$$

$$y_2 = 15 \frac{(1+s)(3+s)}{(16+75)(6+s)} - \frac{k_b s}{(5+6)}$$

$$\frac{1}{\frac{1}{R} + \frac{1}{sC}} = Y = \left(R + \frac{1}{sC}\right)^{-1} \rightarrow \frac{s \frac{1}{R}}{s + \frac{1}{RC}} \quad k_b = \frac{1}{R}$$

$$k_b = \left. y_2 \frac{(s+6)}{s} \right|_{s \rightarrow -6} = 15 \frac{(1-6)(3-6)}{(16-(7 \cdot 6)) - 6} = \frac{15(-5)(-3)}{-26 \cdot -6} = \frac{15^2}{156} = \frac{75}{52}$$

$$RC = \tau = \frac{1}{b} \quad \wedge \quad \frac{1}{R} = \frac{75}{52}$$

$$\frac{75}{52} = c$$

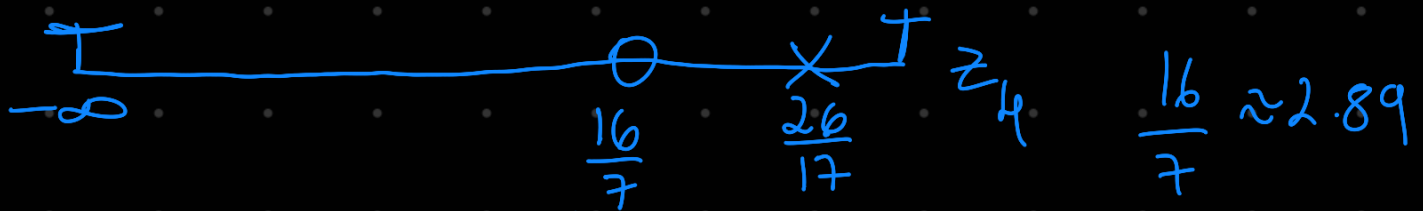
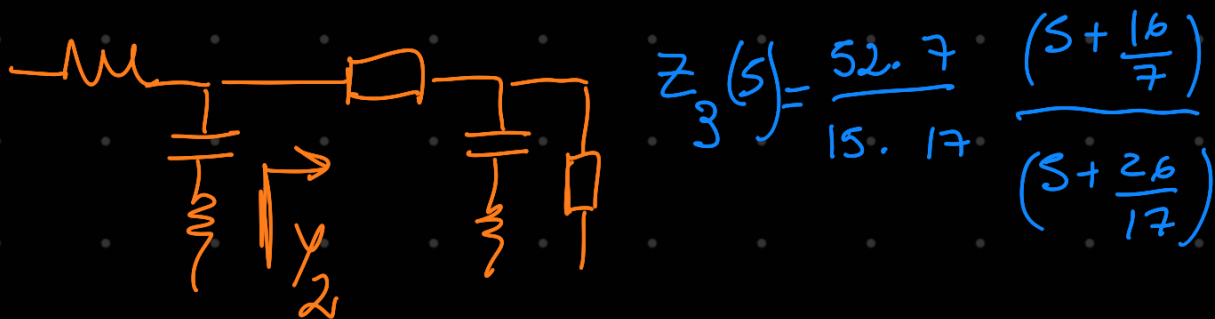
$$\left\{ \begin{array}{l} c = k_b \cdot \tau = \frac{25}{204} \\ R = \frac{52}{75} \end{array} \right.$$

$$y_2 = \frac{(1+s)(3+s) - \frac{75}{52} s (16+75)}{(16+75)(5+6)}$$

$$45 + 60s + 15s^2 - \frac{75}{52} \cdot 16s - \frac{75}{52} \cdot 7s^2$$

$$Y_2 = \frac{\frac{255}{52}s^2 + \frac{480}{13}s + 45}{(16+7s)(s+6)} = \frac{15}{52} \frac{(26+17s)(6+s)}{(16+7s)(6+s)}$$

$$Y_2 = \frac{15}{52} \cdot \left(\frac{26+17s}{16+7s} \right) = \frac{15 \cdot 17 \left(s + \frac{26}{17} \right)}{52 \cdot 7 \left(s + \frac{16}{7} \right)}$$



Busco un cero en $7/2$ así Y_3 tiene un polo y puedo removerlo $\frac{1}{T}$

$$z_4 = z_3(s) - z_3\left(-\frac{7}{2}\right) \quad 884/1005$$

$$z_4 = \frac{52.7}{15 \cdot 17} \frac{\left(s + \frac{16}{7}\right)}{\left(s + \frac{26}{17}\right)} - \frac{52.7}{15 \cdot 17} \frac{\left(-\frac{7}{2} + \frac{16}{7}\right)}{\left(-\frac{7}{2} + \frac{26}{17}\right)}$$

$$\frac{364}{255} \left[\frac{s + \frac{16}{7}}{s + \frac{26}{17}} - \frac{289}{469} \right]$$

Remoción Parcial

$$\frac{884}{1005}$$

$$z_4 = \frac{364}{255} \left[\frac{s + \frac{16}{7} - \frac{289}{469} s - \frac{289 \cdot 26}{469 \cdot 17}}{s + \frac{26}{17}} \right]$$

$$z_4 = \frac{364}{255} \frac{s \cdot 180/469 + \frac{90}{67}}{s + \frac{26}{17}}$$

$$z_4 = \frac{364}{255} \frac{180}{469} \frac{(s + 7/2)}{(s + 26/17)}$$

$$Z_4 = \frac{624}{1139} \frac{(s + 7/2)}{(s + 26/17)} \rightarrow Y_4 = \frac{1139}{624} \frac{s + \frac{26}{17}}{s + 7/2}$$

$$Y_5 = \frac{1139}{624} \frac{s + \frac{26}{17}}{s + 7/2} - \frac{K_{7/2} s}{(s + 7/2)}$$

$$K_{7/2} = \left. Y_4 \frac{(s + 7/2)}{s} \right|_{s \rightarrow -7/2} = \frac{1139}{624} \frac{(-7/2) + \frac{26}{17}}{-7/2} = \frac{4489}{4368}$$

$$RC = Z = \frac{2}{7} \wedge \frac{1}{R} = \frac{4489}{4368} \quad \left\{ \begin{array}{l} C = K_{7/2} \cdot Z = \frac{4489}{15288} \\ R = \frac{4368}{4489} \end{array} \right.$$

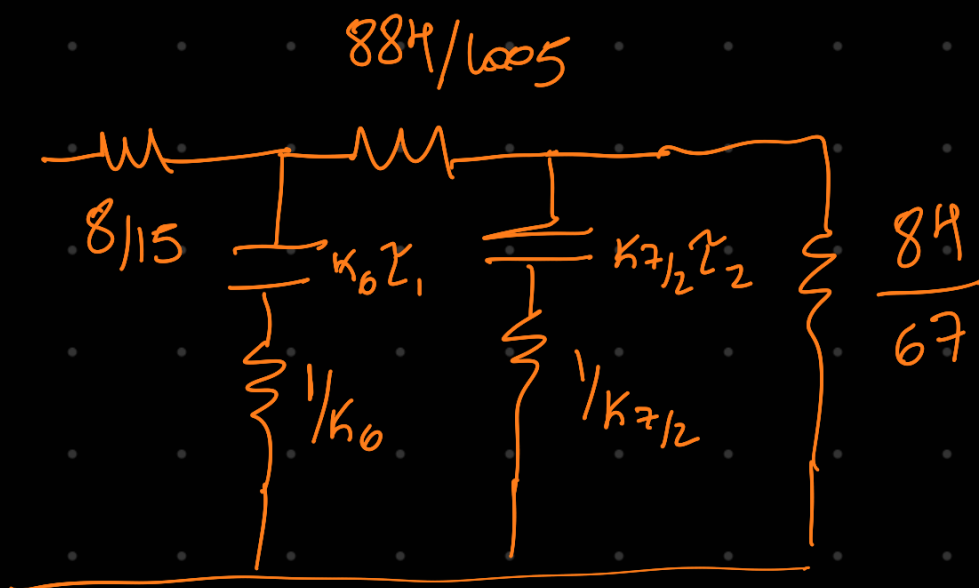
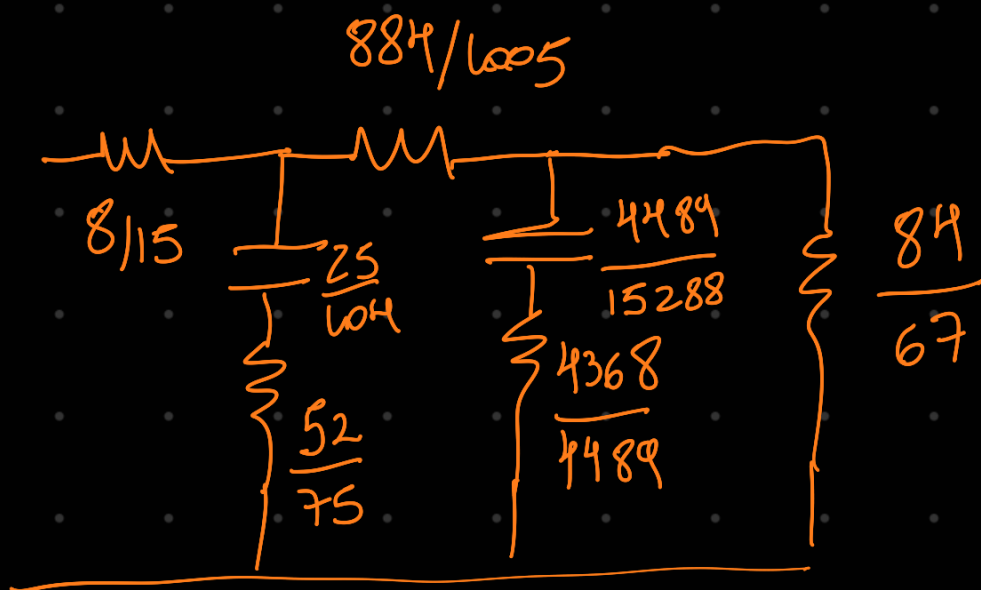
$$C = \frac{4489}{15288}$$

$$Y_5 = \frac{1139}{624} \left(s + \frac{26}{17} \right) - \frac{4489}{4368} s$$

$$(s + 7/2)$$

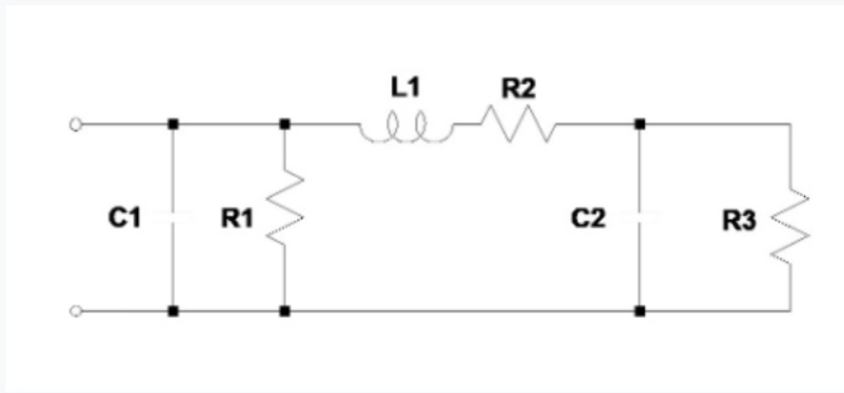
$$Y_5 = \frac{\frac{67}{84} \cdot 5 + \frac{67}{24}}{(5 + 7/2)} = \frac{67}{84} \frac{(5 + \cancel{\frac{7}{2}})}{(\cancel{5 + 7/2})}$$

$$Y_5 = \frac{67}{84} \Rightarrow Z_5 = \frac{84}{67} \quad \text{or} \quad \frac{84}{67}$$



3) Determine el valor de los componentes que integran el siguiente dipolo, sabiendo que satisface la impedancia propuesta:

$$Z(s) = \frac{s^2 + s + 1}{(s^2 + 2s + 5)(s + 1)}$$



$$Y = \frac{(s^2 + 2s + 5)(s + 1)}{s^2 + s + 1} = \frac{s^3 + 3s^2 + 7s + 5}{s^2 + s + 1}$$

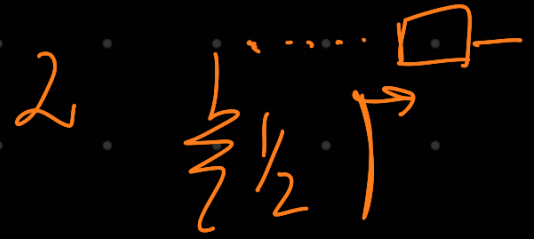
$$\begin{array}{r} s^3 + 3s^2 + 7s + 5 \quad | \quad s^2 + s + 1 \\ \underline{s^3 + s^2 + s} \quad \quad \quad s \\ s^2 + s + 1 \quad | \quad 2s^2 + 6s + 5 \end{array}$$

$$Z_2 = \frac{s^2 + s + 1}{2s^2 + 6s + 5} \rightarrow Y_2 = \frac{s^2 + s + 1}{2s^2 + 6s + 5}$$

$$\begin{array}{r} 2s^2 + 6s + 5 \\ - (2s^2 + 2s + 2) \\ \hline \end{array}$$

$$s^2 + s + 1 \quad \overline{4s + 3}$$

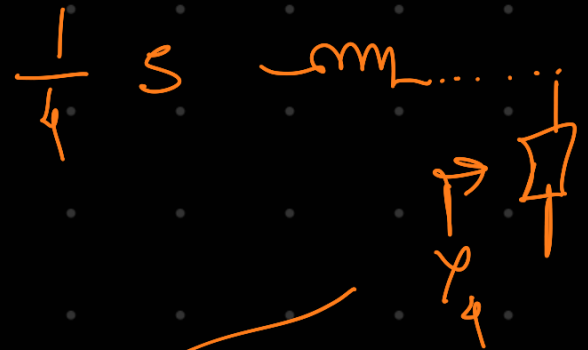
$$Z_3 = \frac{s^2 + s + 1}{4s + 3}$$



$$\begin{array}{r} s^2 + s + 1 \\ - (s^2 + \frac{3s}{4}) \\ \hline \end{array}$$

$$4s + 3 \quad \overline{\frac{1}{4}s + 1}$$

$$\overline{4s + 3}$$



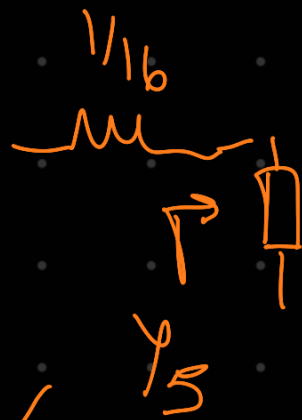
$$\cancel{Y_4} = \frac{4s + 3}{\frac{1}{4}s + 1} \rightarrow Z_4 = \frac{\frac{1}{4}s + 1}{4s + 3}$$

$$\frac{1}{4} s + 1$$

$$\frac{4s+3}{16}$$

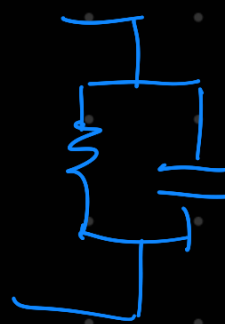
$$\frac{1}{4} s + \frac{3}{16}$$

$$\frac{1}{4.4} = \frac{1}{16}$$



$$\frac{4s+3}{16}$$

$$Y_5 = \frac{4s+3}{16}$$



$$Y = \left(\frac{1}{R} + sC \right)$$

$$= \frac{1 + sCR}{R}$$

$$= \frac{\left(\frac{1}{R_C} + s \right) R_C}{R_C}$$

$$Y_5 = \left(s + \frac{3}{4} \right) \frac{16 \cdot 4}{13}$$

$$\frac{1}{R_C} = \frac{3}{4}$$

$$C = \frac{16 \cdot 4}{13} = \frac{64}{13}$$

$$R_C = \frac{4}{3} \rightarrow R = \frac{4}{3} \cdot \frac{1}{C} = \frac{13}{48}$$

